LIFE OS

Case Studies

Reality grounding and conceptual stress tests - Checkpoint v0.1

Purpose Preservare i casi reali che hanno generato il modello, cosi che le future astrazioni possano essere verificate contro comportamenti, frizioni e bisogni concreti.

Status	Foundation / evidence corpus
Reading mode	Reality grounding / stress tests
Canonical sources	Master Extraction v0.1 §§60-74, §107
Cases	Solducci, LaGenteStrana, Agrivoltaico, HistoryExplorer, Speedcubing, Crochet

Case studies are not implementation specifications. They preserve the lived problem before theory and identify what future ontology must be able to represent.

0. How to use the case studies

Questi casi funzionano come integration tests concettuali: quando un modello cambia, deve essere possibile tornare qui e verificare se il comportamento reale resta rappresentabile senza boilerplate o special case arbitrari.

- Preservare il comportamento naturale prima della formalizzazione.
- Separare frizione reale da soluzione ipotizzata.
- Registrare i concetti emersi dal caso, senza trasformarli automaticamente in primitive.
- Mantenere anche cio che si e deciso di non modellare.
- Usare le stress-test questions per Ontology Minimum v0.1 e versioni successive.

0.1 Coverage and current gaps

Strong coverage	Solducci, Speedcubing
Good coverage	Crochet, Agrivoltaico
Medium coverage	LaGenteStrana, HistoryExplorer
Future second-wave tests	Health, Relationships, Bureaucracy, Travel, Recurring maintenance

CS01 - Solducci

Da shared-expense tracker a piattaforma per la gestione di risorse condivise; caso principale per Project, Idea, Knowledge contestuale e visione progressiva.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Nasce da un bisogno reale e immediato, poi espande la propria identita nel tempo.
- Combina utilita personale, design di prodotto, imprenditorialita e apprendimento tecnico/normativo.
- Produce molti livelli di struttura: visione, feature, open points, idee, knowledge, implementation e reusable patterns.

2. Natural behavior before the system

- Origine in un Google Sheet condiviso per tracciare spese con la partner.
- Trasformazione progressiva in app e poi in idea piu ampia di gruppi e spazi condivisi.
- Le nuove feature emergono da bisogni, brainstorming, sinergie e osservazione di come il prodotto potrebbe ridurre boilerplate nella vita.
- Le idee vengono annotate con granularita variabile e possono macerare prima di entrare in backlog.

3. Original friction

- Visione ampia con idee sparse e componenti temporanei che cambiano forma.
- Knowledge generica (mobile app, GDPR, security, publishing) si intreccia con blocker specifici di Solducci.
- Difficolta nel vedere contemporaneamente stato consolidato, open points, next steps e future possibilities.
- Rischio di duplicare conoscenza generica dentro ogni progetto.

4. Desired support

- Project vitrine con identita, fase corrente, milestone, stato consolidato e next actions.
- Visione componibile in aree/feature che possano apparire, fondersi o essere archiviate senza perdere memoria strutturale.
- Collegare generic Knowledge e mostrare, con gli occhi di Solducci, quali parti sono rilevanti o blocker.
- Idea Inbox con crescita progressiva delle idee e promozione volontaria quando diventano abbastanza concrete.
- Post-implementation comparison tra aspettative, realta e requisiti ancora non implementati.

5. Information and entities involved

- Project: Solducci.
- Vision/Product components: Financial Hub, Spaces, Notes, Pantry, Shopping, Tasks, Canvas, Bento Dashboard, Context Manager.
- Generic Concepts: App Mobile, GDPR, authentication, privacy/data handling, publishing standards, backend/resource distribution.
- Ideas con body, sub-ideas, screenshots, requirements, advantages, KPI links e feature influence.
- Technical choices and design patterns come knowledge/progetto riutilizzabile.

6. Processes and activities

- Feature ideation tramite domande ricorrenti: cosa voglio ottenere, come, quale interazione umana, come integra, quali sinergie.
- Exploration di open points e knowledge gaps.
- Execution attraverso backlog, milestone, task e development sessions.
- Retrospective/post-mortem su implementazione vs aspettative.
- Extraction di reusable knowledge e pattern.

6.1 Current vision sample

- Financial Area / Hub.

- Spaces come contenitore temporaneo di Notes, Pantry, Shopping Lists e Tasks.
- Canvas come note-file system.
- Bento widget homescreen.
- Context Manager come infrastruttura di contesto selezionato persona/gruppo.

7. Concepts this case helped discover

- Project come Entity con identity, evolution, operational memory e progressive decomposition.
- Generic Knowledge separata dal contextual state: GDPR non e globalmente blocker; Solducci-GDPR puo esserlo.
- Relation metadata per relevance, blocker, addressed/partial.
- Progressive structuring e Entity promotion.
- Structural history: preservare la struttura passata senza snapshot completi del contenuto.
- Goal/progress relativo alla visione corrente.

8. Deliberately not modeled

- Snapshot completi di ogni versione del contenuto.
- Schema rigido e definitivo delle feature.
- Duplicazione di tutte le regole mobile dentro Solducci.
- Promozione automatica delle idee a Project/Feature.

9. Ontology stress-test questions

- Un Entity/Relation model minimo riesce a rappresentare App Mobile come conoscenza generica e Solducci come caso concreto senza duplicazione?
- Una feature temporanea puo essere riorganizzata mantenendo una memoria strutturale utile?
- Goal, Scenario, Milestone e Phase possono essere distinti senza introdurre troppi tipi?
- Relation metadata e sufficiente per rappresentare relevance, blocker e partial coverage?

CS02 - LaGenteStrana

AI software-development tool costruito con un amico stretto; caso per collaborazione, progetto ad alto valore e requirements che emergono dall uso.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Rappresenta la direzione imprenditoriale e il desiderio di aumentare autonomia attraverso automazione.
- E collaborativo e nasce da pratiche gia esistenti, non da un idea astratta isolata.
- Il tool puo essere usato per scoprire le proprie funzionalita mancanti, creando un loop progetto-strumento.

2. Natural behavior before the system

- Entrambi i collaboratori automatizzavano separatamente workflow software.
- Condivisione di scoperte, brainstorming e repository fino all emergere di un vero Project.
- La priorita attuale e raggiungere un punto realmente utilizzabile.
- Molte idee rimangono sparse e vengono valutate attraverso il valore che aggiungono all uso concreto.

3. Original friction

- Idee eterogenee e evolving requirements.
- Need to preserve shared context without forcing excessive project administration.
- Learning e execution si alimentano reciprocamente.
- Priorita alta ma molte possibili direzioni tecniche/prodotto.

4. Desired support

- Resume chiaro del progetto e current missing functionality.
- Milestone e next actions vicino al presente, visione piu astratta sul futuro.
- Collegamenti con knowledge e lessons utili anche ad altri software projects.
- Rappresentazione del contributo collaborativo senza trasformare il Life OS in project-management enterprise.

5. Information and entities involved

- Project, collaborators, repositories, ideas, features, technical knowledge, blockers, usage observations.
- Potential relations to other projects that can learn from LaGenteStrana.

6. Processes and activities

- Collaborative exploration.
- Dogfooding / use-driven discovery.
- Execution and iteration.
- Knowledge extraction and transfer.

7. Concepts this case helped discover

- Project identity can emerge from repeated activity and shared commitment.
- System should surface synergies between projects.
- Current phase and missing functionality are valuable resume signals.

8. Deliberately not modeled

- Fine-grained permissions, ownership or enterprise collaboration model at this checkpoint.
- Full issue-tracker replacement.

9. Ontology stress-test questions

- Can the ontology represent collaboration lightly, without making every relationship a workflow object?
- Can observations generated while using the tool become ideas or knowledge without forced typing?

- Can project-to-project learning be represented as relations rather than duplication?

CS03 - Agrivoltaico

Idea imprenditoriale ad alta incertezza e alto commitment potenziale; caso principale per Exploration, Scenario, knowledge gaps e parking.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Dimostra che una cosa puo essere importante per anni senza essere attivamente eseguita.
- Richiede competenze lontane dal dominio professionale corrente e investimenti rilevanti.
- La decisione di commitment dovrebbe richiedere piu evidenza e reality check rispetto a un hobby economico.

2. Natural behavior before the system

- Pensiero intermittente per mesi/anni, con poche note e molta elaborazione mentale.
- L idea ritorna per attrazione verso sostenibilita, profitto, scalabilita e accessibilita.
- Viene spesso deprioritizzata per mancanza di competenze, capitale e spazio mentale.

3. Original friction

- Manca una visione completa delle parti da comprendere.
- Knowledge gaps non sono visibili in modo strutturato.
- L inattivita puo far sembrare il progetto dimenticato anche quando resta significativo.
- Rischio di studiare casualmente senza raggiungere una soglia di decisione.

4. Desired support

- Exploration view con domande, unknowns, resource paths e scenarios.
- Macro decomposition: agricoltura, fotovoltaico, distribuzione energia, terreno, finanziamento.
- Commitment threshold esplicito: quali informazioni/condizioni servono prima di investire seriamente.
- Parking che preservi il quadro e faccia emergere il tema quando torna rilevante.

5. Information and entities involved

- Idea/Project candidate, scenarios, domain concepts, resources, financial assumptions, land, regulation, energy distribution, risks, required skills.

6. Processes and activities

- Exploration.
- Scenario comparison.
- Feasibility study.
- Knowledge acquisition.
- Long-term parking and periodic review.

7. Concepts this case helped discover

- Exploration as mode/process rather than universal Entity.
- Scenario as alternative high-level configuration.
- Large commitments require stronger evidence/reality checks.
- Importance-weighted inactivity.

8. Deliberately not modeled

- Hypothesis or Evidence as mandatory universal types.
- Detailed financial model before the project reaches that maturity.
- Automatic commitment based on recurring interest.

9. Ontology stress-test questions

- Can an entity remain in Exploration for years without becoming a Project full of empty fields?
- Can scenarios share substructures rather than duplicate entire trees?
- Can parking preserve significance while avoiding noisy reminders?
- Can required knowledge be shown as contextual gaps without contaminating generic concepts?

CS04 - HistoryExplorer

Interactive history exploration tool with a history-enthusiast friend; case for low-intensity collaboration, delegation and synergy.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Shows a project that is meaningful but intentionally low effort.
- Value includes fun, learning and niche empowerment, not only entrepreneurship.
- May be handed off after initial setup, testing whether the model tolerates asymmetric commitment.

2. Natural behavior before the system

- One brainstorming session generated ideas and wireframes.
- Potential plan: create the initial setup and let the friend carry more development.
- Can generate learning useful for LaGenteStrana.

3. Original friction

- Too much formal project management would cost more than the project deserves.
- Need to preserve brainstorming and resumption context despite low activity.
- Need to express that ownership/effort may differ between collaborators.

4. Desired support

- Very light Project representation.
- Clear current state and next enabling action.
- Links to transferable knowledge and other projects.
- Ability to remain dormant without appearing broken.

5. Information and entities involved

- Project, collaborator, wireframes, ideas, initial setup, learning links.

6. Processes and activities

- Brainstorming/exploration.
- Light execution.
- Delegation/handoff.
- Knowledge transfer.

7. Concepts this case helped discover

- Complexity must be proportional to commitment and maturity.
- A Project can be real even with low cadence.
- Synergy is a first-class system responsibility even if not a first-class ontology type.

8. Deliberately not modeled

- Detailed ownership model.
- Heavy recurring ceremony specific to the project.

9. Ontology stress-test questions

- Can Project exist with minimal schema and still be resumable?
- Can low cadence be distinguished from accidental neglect?
- Can project synergy be derived from shared knowledge/relations?

CS05 - Speedcubing

Practice domain used to distinguish skill progression, sessions, metrics, observations and knowledge from project completion.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Introduces repeated practice rather than a finite artifact-focused Project.
- Performance can be measured, but measurement must remain subordinate to learning and awareness.
- Combines theory, algorithms, motor technique, free solves and reflective practice.

2. Natural behavior before the system

- Learned layer method and part of CFOP/Friedrich.
- Studies videos/articles to understand possible improvement paths before deciding what to train.
- Mixed sessions: timed solves, slow reflective solves, algorithm learning/review, finger movement practice, news/content.
- Milestones can overlap and be reordered based on observed bottlenecks.

3. Original friction

- cstimer stores solve data but not conveniently the qualitative observations that matter.
- Tracking every solve in Life OS would be redundant and high-maintenance.
- Practice quantity is useful to understand attention distribution but is not the value of the goal.
- Need to resume what algorithms are learned/learning and what current bottleneck is.

4. Desired support

- Where am I view: current phase, milestones, algorithms learned/learning, trends and recent practice.
- What can I do view: task, review knowledge, read, timed solve, slow solve, practice algorithm.
- Session-level metrics such as Single, Ao5, Ao12 plus optional observations/sensations.
- Retrospectives that aggregate recurring bottlenecks, e.g. how often F2L was identified as limiting.

5. Information and entities involved

- Practice, Session, Metric, Observation, algorithms, solving methods, F2L/OLL/PLL Concepts, milestones, resources.

6. Processes and activities

- Practice consisting of heterogeneous sessions.
- Study.
- Performance observation.
- Retrospective and milestone reprioritization.

6.1 Current milestone sample

- All PLL reasonably well.
- Most OLL.
- Finger tricks/movements.
- Improve cube perception / lookahead before moves.
- Train F2L and review algorithms.

7. Concepts this case helped discover

- Session as a sample of lived activity, not exhaustive event logging.
- Metric as structured measure of reality associated with a Session.
- Observation as optional reflective property/content, often attached to sessions.
- Practice as process over multiple sessions.

- Knowledge completion can be derived from subparts.
- Do not model method-per-session until multiple methods make the distinction useful.

8. Deliberately not modeled

- Every individual solve in Life OS.
- Mandatory comments after every session.
- Hypothesis and Evidence as formal types.
- Future timing metrics per individual algorithm before they are needed.

9. Ontology stress-test questions

- Can the ontology represent Practice without forcing it into Project completion semantics?
- Can metrics be optional and session-scoped?
- Can F2L be one Concept reused across solving-method views?
- Can knowledge/progress be derived from algorithm-level state without redundant percentages?

CS06 - Crochet

Creative case that connects inspiration, Idea, progressive structure, Project execution, Artifact and reusable Knowledge.

WORKING	Reality-grounding case study. The representation below preserves observed behavior before final ontology formalization.

1. Why this case matters

- Shows how inspiration can remain a simple connection rather than dedicated type.
- Demonstrates a natural transformation from vague creative intent to executable Project.
- Shows that a finished Project can be dehydrated into reusable pattern/knowledge.

2. Natural behavior before the system

- Pinterest collections store visual references by hobby.
- When an idea appears, a lightweight note is created and linked to relevant inspiration.
- Generic crochet knowledge contains base structures/methods, stitch types, steps and intermediate photos.
- An idea can reference known structures inline, e.g. square -> cone -> triangle.
- Once vision is clear and time is available, the idea can be transformed into an active Project.

3. Original friction

- Active creative projects can be forgotten between bursts of attention.
- Ideas on paper or image boards are disconnected from executable structure.
- Repeated learning may remain trapped inside one project instead of becoming a reusable pattern.
- A rigid project template would be disproportionate while the idea is still only a sketch.

4. Desired support

- Very cheap capture of Idea and inspiration links.
- Progressive decomposition into construction steps, materials and tasks only when needed.
- Sessions/comments/photos during execution.
- After completion, easy extraction of generalized pattern stripped of project-specific noise.

5. Information and entities involved

- Resource/image, Idea, crochet Concepts, structure/pattern, Project, Task, Session, progress photo, Artifact, Observation, reusable Knowledge.

6. Processes and activities

- Inspiration/connection.
- Idea development and maceration.
- Exploration of techniques.
- Project execution.
- Practice.
- Artifact creation.
- Knowledge extraction/generalization.

7. Concepts this case helped discover

- Inspiration need not be a first-class Entity type.
- Same object can mature rather than being copied through rigid workflow states.
- Activities generate information: artifact, observation, knowledge, idea.
- Specific mistakes can remain observations; repeated/generalized lessons can become Knowledge.
- Past work should reduce future work.

8. Deliberately not modeled

- Mandatory Inspiration type.

- Forced workflow Inspiration -> Idea -> Project -> Pattern.
- Automatic conversion of every project observation into Knowledge.

9. Ontology stress-test questions

- Can a raw image/resource connect to an Idea without special-case ontology?
- Can a simple Idea acquire Project capabilities without duplication?
- Can an Artifact be represented as a normal Entity produced-by activity/project?
- Can generalized pattern knowledge reference current Concepts while preserving useful project history?

10. Cross-case synthesis

I casi non sono sei esempi dello stesso workflow. Il loro valore e mostrare quali meccanismi devono restare generali e quali differenze non devono essere appiattite.

Dimension	Solducci	LaGenteStrana	Agrivoltaico	HistoryExplorer	Speedcubing	Crochet
Project identity	Strong	Strong	Candidate	Light	Weak	Strong when committed
Exploration	Medium	Medium	Strong	Medium	Medium	Medium
Practice / Session	Medium	Medium	Low	Low	Strong	Strong
Generic Knowledge	Strong	Strong	Strong	Medium	Strong	Strong
Metrics	Low	Low	Potential	Low	Strong	Optional
Scenario	Medium	Low	Strong	Low	Low	Low
Resume after inactivity	Strong	Strong	Strong	Strong	Strong	Strong
Progressive structuring	Strong	Strong	Strong	Strong	Strong	Strong

10.1 Repeated design evidence

- Resume value is universal across active and dormant cases.
- Progressive structuring appears in projects, knowledge and creative ideas.
- A small set of reusable relations is preferable to domain-specific duplicated structures.
- Session/Practice are necessary for repeated activity but not for every Project.
- Metrics are useful in some domains and unnecessary in others; optionality is essential.
- Contextual state must stay separate from generic knowledge.
- Low-commitment cases are an important guard against over-engineering.

10.2 Missing test domains

Coverage gap The current evidence corpus is strong on Projects, Knowledge and Practice, but weaker on Health, Relationships, Bureaucracy, Travel and recurring maintenance. These should become second-wave stress tests before considering the ontology broadly stable.

- Health: recurring checks, uncertain symptoms, appointments, private notes and maintenance without turning health into a project.
- Relationships: long-lived bonds, shared moments, maintenance, reconnection and subjective quality without scoring people.
- Bureaucracy/finance: recurring obligations, deadlines, documents and low-interest but high-importance tasks.
- Travel: temporary project-like logistics plus experience sampling.
- Recurring maintenance: cadence, expected frequency and debt without habit-tracker rigidity.

11. Case-study acceptance rule for Ontology Minimum v0.1

Acceptance rule A candidate ontology is not successful because it can encode these cases in principle. It should represent them naturally, with low boilerplate, progressive depth, correct contextuality and no proliferation of special-case types.

- Represent the strong cases without losing behavior that motivated the model.
- Keep simple cases simple.
- Allow richer cases to deepen without migrating to a different universe of objects.

- Preserve optionality of metrics, observations and processes.
- Make resumability and reuse easier than in the original behavior.
- Keep open questions open when evidence is insufficient.